

CoARA Action Plan



TECHNISCHE
UNIVERSITÄT
DARMSTADT

2024 – 2027

About TU Darmstadt

[TU Darmstadt](#) is one of Germany's leading technical universities and a synonym for excellent, relevant research. Global transformations – from the energy transition via Industry 4.0 to artificial intelligence – are challenges TU Darmstadt embraces wholeheartedly. We are crucially shaping these far-reaching processes of change with outstanding insights and forward-looking study opportunities.

TU Darmstadt pools its cutting-edge research in three fields: Energy and Environment, Information and Intelligence, Matter and Materials. Our problem-based interdisciplinarity involving the engineering sciences, natural sciences, humanities, and social sciences as well as our productive interaction with society, business and politics generate progress towards sustainable development worldwide.

Since we were founded in 1877, we have been one of Germany's most international universities; as a European technical university, we are a driving force in developing a trans-European campus together with universities from nine European countries in the network, [Unite!](#) Our home is the metropolitan region Frankfurt-Rhine-Main. With our partners, the [Rhine-Main universities](#) – Goethe University Frankfurt and Johannes Gutenberg University Mainz – we further the development of this globally attractive science location.

Coalition for Advancing Research Assessment (CoARA)

In November 2022, TU Darmstadt signed the Agreement on Reforming Research Assessment (ARRA) and joined the Coalition for Advancing Research Assessment (CoARA) from its inception. As part of the Rhine-Main university alliance (RMU), which is also a CoARA member, TU Darmstadt is an active member of the CoARA working group "Reforming Academic Career Assessment (ACA)". In March 2023, TU Darmstadt joined the newly founded CoARA National Chapter Germany.

Focus areas

By signing the Agreement on Reforming Research Assessment, TU Darmstadt has acknowledged the Core and Supporting Commitments and initiated an internal reflection process on promoting responsible research assessment. While the implementation of the commitments concerns the university as a whole, three focus areas were selected, which will strongly benefit from the reform process. In addition, various other processes will be reviewed and revised during the timeframe of this action plan.

Early Career Researcher

TU Darmstadt pursues a consistent concept in the promotion of early career researchers, structured along qualification phases: prospective doctoral candidates, doctoral phase (R1), postdoctoral phase (R2), and qualification phase for a professorship (R3). In all qualification phases, the University aims to create optimal framework conditions and offer an outstanding environment in which young researchers can develop their potential to the best of their ability.

[Ingenium – Young Researchers at TU Darmstadt](#) serves as umbrella organisation for promoting early career researchers at TU Darmstadt, offering information, guidance, and support in pursuing academic as well as non-academic career paths. As a single point of contact for doctoral candidates and postdoctoral researchers, Ingenium provides a professional skills programme and regularly organises career and networking events.

In the doctoral phase, TU Darmstadt promotes the acquisition of competences for early career researchers through a large number of quality-assured and demand-oriented professional skills

training courses within the framework of the Ingenium Qualification Programme. In addition, the Ingenium Networking Programme offers career-relevant information and orientation events.

Early career researchers in the R2 phase benefit from the Postdoc Career Programmes at TU Darmstadt, which include several support and funding programmes. All of these programmes are designed to enhance career prospects and secure scientific independence. In upcoming funding rounds, we plan to review and revise the selection processes where required to better acknowledge the variety of contributions to and careers in research. Additionally, qualitative criteria in the selection processes will be strengthened and the reliance on quantitative indicators reduced.

A variety of options and career paths are available to scientists in the qualification phase for an unlimited professorship (R3), including appointments as junior research group leaders. With a particular focus on this target group, we plan to design and offer specific services that contribute to knowledge of and sensitisation to the diversity of contributions to and careers in research. It is our intention to incorporate the results and tools from various CoARA working groups.

Professorial Appointments

Professorial appointments represent a pivotal instrument for the academic advancement of the university and are of paramount importance for the reputation of TU Darmstadt. Particularly tenure-track professorships are understood as an attractive and future-oriented career path for early phase researchers since they improve predictability and transparency. At the same time, tenure-track professorships offer an opportunity for departments to attract the leading minds in innovative as well as unconventional research fields. The objective of all professorial appointments at TU Darmstadt – and in particular of the tenure-track professorships – is to attract more women and international researchers, thus making its professorial staff more diverse.

The appointment procedures at TU Darmstadt are typically organised according to a sequence of predefined steps. The university aims for transparent, reliable, and quality-assured processes. Prospective candidates can find all the information they require on the steps in the appointment procedure as well as open positions on the university's website.

TU Darmstadt has recently revised its guidelines on the establishment of assistant professorships. The revised guidelines include clearly defined and transparent criteria for the evaluation of the candidates within three evaluation categories: research, teaching, and leadership. The mandatory template is the same for all tenure-track professorships but allows for the individual design of indicators that signal positive target achievement. With the revision of the tenure-track guidelines and the new indicator-based agreement on objectives, TU Darmstadt is attempting to move away from an understanding of the agreement on objectives as a checklist and to promote the development of the assistant professor's potential during the tenure phase.

As part of an upcoming revision of the general appointment statutes of all professorships (not tenure-track only), the topic of appointment procedures, their quality and best practices – e.g. regarding diversity or the evaluation of research and teaching – will be discussed across the university. It is the intention of the university that journal- and publication-based metrics will only be used in a responsible manner in appointment procedures and regular assessment of performance.

Integrated Quality Management

At TU Darmstadt, the integrated quality management aims to close quality cycles not only at the level of the individual instruments, but also to link the quality assurance measures directly with strategic decisions at departmental and university level. Quality assurance instruments in the areas of research, studies and teaching, early career researchers, management and administration can be located at the levels of activities, programmes, and structures. The university's integrated quality management aims

to link quality assurance and strategic processes at all levels and create synergies. It involves all areas, units, and status groups in the quality assurance processes through a participatory approach and relies on deliberative negotiation processes to establish a common and shared understanding of quality.

The institutional evaluation is the central element of the university's integrated quality management. It covers research and professorial appointments, study and teaching including the development of degree programmes, a maximum of two department-specific topics as well as interdisciplinarity, internationalisation and equal opportunities in all the above-mentioned areas. The evaluation process consists of four consecutive stages: 1) self-evaluation and report, 2) review by an external evaluation committee, 3) agreement on objectives between the Executive Board and the evaluated unit, and 4) implementation and monitoring. This cycle is repeated every six to eight years for each unit.

The self-evaluation report enables each department/unit to showcase its accomplishments and particularly its future plannings in an autonomous manner without relying on predefined metrics or indicators. The second stage, which has a peer-review character, is of special importance. During the on-site visit, an external evaluation committee will discuss the topics covered in the self-evaluation report with representatives from all status groups as well as internal and external cooperation partners. Subsequently, the committee presents a summary of its recommendations for the development of the department and its degree programmes in an evaluation report. This report serves as the basis for an agreement on objectives between the Executive Board and the evaluated unit.

All departments and units have completed at least one or two evaluation cycles. Unlike in previous cycles, the new agreements on objectives will no longer contain research related objectives with bibliometric indicators such as the number of publications in journals with a specific impact factor.

Milestone plan 2024 – 2027

Commitments and Scope	Planned Actions	Timeframe
Commit resources to reforming research assessment as is needed to achieve the organisational changes committed to.	The executive board, in particular the Vice-Presidents responsible for Research and Academic Careers, leads and coordinates all actions planned to implement the CoARA commitments at TU Darmstadt, with the aim of promoting responsible research assessment.	ongoing
Involve your institutional community in the change process.	Information and discussion on the reform process at TU Darmstadt in the university senate.	2024 – 25
Exchange practices and experiences to enable mutual learning within and beyond the Coalition.	- University-wide workshop on the Agreement, its implications and implementation at TU Darmstadt - Exchange experiences internationally through Coalition workshops and active participation in the CoARA working group ACA. As a member of the German National Chapter, we discuss and share our progress with other members and interested organisations.	ongoing
Review and develop research assessment criteria, tools, and processes.	<u>Early Career Researcher</u> Review and revise the information services/workshops for early career researcher (R1 to R3) contributing to an awareness of and sensitisation to the diversity of contributions to research and scientific careers.	2024 – 25
Recognise the diversity of contributions to, and careers in, research in accordance with the needs and nature of the research.	Design of specific services, which contribute to knowledge of and sensitisation to the diversity of contributions to and careers in research, for particular target groups (R3 researcher, i.e. Athene Young Investigators, assistant professors, junior research group leaders - but also research mentors, participants in selection and appointment committees).	2025 – 26
Base research assessment primarily on qualitative evaluation for which peer review is central, supported by responsible use of quantitative indicators.	Review and revise the selection processes for TU Darmstadt's funding formats for R2 and R3 researchers (mobility grant, career bridging grant, Athene Young Investigator) to better acknowledge the variety of contributions to and careers in research. Additionally, strengthen qualitative criteria and reduce the reliance on quantitative indicators.	2025 – 26

Commitments and Scope	Planned Actions	Timeframe
Review and develop research assessment criteria, tools, and processes. Recognise the diversity of contributions to, and careers in, research in accordance with the needs and nature of the research. Base research assessment primarily on qualitative evaluation for which peer review is central, supported by responsible use of quantitative indicators.	<u>Professorial Appointments</u> – Revision of the publicly available guidelines for assistant and tenure-track professorships to ensure a transparent, reliable and quality assured process. The evaluation is based on clearly defined and transparent criteria. The individual target criteria fall into the categories of research, teaching, and leadership, allowing for subject-specific and research career-specific agreements. – Revision of the general appointment statutes defining the future role of h-indices and Journal-impact factors in appointment procedures and performance assessments. <u>Integrated Quality Management</u> – Peer review by an external evaluation committee plays a central role in the institutional evaluation of departments, central services, and central administrations. The recommendations of the committee on the development of the department and its degree programmes serve as a basis for an agreement on objectives between the Executive Board and the evaluated unit.	2024 2025 – 27 ongoing
Abandon inappropriate uses in research assessment of journal- and publication-based metrics, in particular inappropriate uses of Journal Impact Factor (JIF) and h-index.	– Exclude the development of the institutional h-index as well as a comparison with h-indexes of other universities from TU Darmstadt's internal reporting system. – Within the institutional evaluation, all upcoming agreements on objectives with TU Darmstadt's departments will no longer contain objectives including Journal Impact Factors and h-indexes.	2024 ongoing
Avoid the use of rankings of research organisations in research assessment.	– TU Darmstadt is actively involved in the service project "International University Rankings" of the German Rectors' Conference (HRK), which scrutinises and discusses the methods and data requests of ranking providers, but also seeks a critical exchange with them. – Until 2024, all of TU Darmstadt's ranking results were published in a ranking report on the university's website. This report will be discontinued. – Start a dialogue on the usage of the results from different ranking organisations in various areas of university such as marketing, international student programmes, research applications.	ongoing starting 2024 starting 2024

Commitments and Scope	Planned Actions	Timeframe
Raise awareness of research assessment reform and provide transparent communication, guidance, and training on assessment criteria and processes as well as their use.	- The different workshop programmes at TU Darmstadt and the Rhine-Main Universities, e.g. by Ingenium, the Human Resources and Organisational Development unit, the Gender Equality Office, or the Diversity Education Office, offer information and workshops on various topics such as bibliometrics and indicators, gender bias, career development.	ongoing